Curated pin names for your own design. No regulator, clock, reset or power-control circuits.
Connector pad numbering follows the workbook, not a verified physical footprint.
Read docs/kicad.md and docs/sources.md before reuse.

Connector A — 400 contacts



File: Connector_A.kicad_sch

Connector B — 400 contacts



File: Connector_B.kicad_sch

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Sheet: /
File: V100_Connectors.kicad_sch

Title: V100 SXM2 connector-only starter

Size: A4	Date: 2026-10-29	**Rev: Private review**
KiCad E.D.A. 10.0.7		Id: 1/3

All pins are exposed for reuse. Unconnected here does not mean an official NC assignment.

J1A

A1	GND
A2	PERp0
A3	PERn0
A4	GND
A5	PERp7
A6	PERn7
A7	GND
A8	PERp8
A9	PERn8
A10	GND
A11	PERp12
A12	PERn12
A13	GND
A14	GND
A15	UNRESOLVED/NC
A16	UNRESOLVED/NC
A17	GND
A18	PWR_GOOD
A19	GND
A20	+5V_AUX
A21	GND
A22	12V
A23	12V
A24	GND
A25	12V
A26	12V
A27	GND
A28	12V
A29	12V
A30	GND
A31	12V
A32	12V
A33	GND
A34	12V
A35	12V
A36	GND
A37	12V
A38	12V
A39	GND
A40	12V

Column A

J1B

B1	PERp1
B2	PERn1
B3	GND
B4	PERp4
B5	PERn4
B6	GND
B7	GND
B8	GND
B9	PERp9
B10	PERn9
B11	GND
B12	PERp15
B13	PERn15
B14	GND
B15	GND
B16	UNRESOLVED/NC
B17	UNRESOLVED/NC
B18	GND
B19	GND
B20	+5V_AUX
B21	GND
B22	12V
B23	12V
B24	GND
B25	12V
B26	12V
B27	GND
B28	12V
B29	12V
B30	GND
B31	12V
B32	12V
B33	GND
B34	12V
B35	12V
B36	GND
B37	12V
B38	12V
B39	GND
B40	12V

Column B

J1C

C1	GND
C2	PERp3
C3	PERn3
C4	GND
C5	PERp6
C6	PERn6
C7	GND
C8	PERp10
C9	PERn10
C10	GND
C11	PERp13
C12	PERn13
C13	GND
C14	GND
C15	UNRESOLVED/NC
C16	UNRESOLVED/NC
C17	GND
C18	I2C_ADDR0
C19	GND
C20	+5V_AUX
C21	GND
C22	12V
C23	12V
C24	GND
C25	12V
C26	12V
C27	GND
C28	12V
C29	12V
C30	GND
C31	12V
C32	12V
C33	GND
C34	12V
C35	12V
C36	GND
C37	12V
C38	12V
C39	GND
C40	12V

Column C

J1D

D1	PERp2
D2	PERn2
D3	GND
D4	PERp5
D5	PERn5
D6	GND
D7	GND
D8	GND
D9	PERp11
D10	PERn11
D11	GND
D12	PERp14
D13	PERn14
D14	GND
D15	GND
D16	UNRESOLVED/NC
D17	UNRESOLVED/NC
D18	GND
D19	GND
D20	+5V_AUX
D21	GND
D22	12V
D23	12V
D24	GND
D25	12V
D26	12V
D27	GND
D28	12V
D29	12V
D30	GND
D31	12V
D32	12V
D33	GND
D34	12V
D35	12V
D36	GND
D37	12V
D38	12V
D39	GND
D40	12V

Column D

J1E

E1	GND
E2	GND
E3	GND
E4	GND
E5	GND
E6	GND
E7	PCIE_REFCLK_P
E8	GND
E9	GND
E10	GND
E11	GND
E12	GND
E13	GND
E14	GND
E15	GND
E16	GND
E17	GND
E18	PERST#
E19	GND
E20	+5V_AUX
E21	GND
E22	12V
E23	12V
E24	GND
E25	12V
E26	12V
E27	GND
E28	12V
E29	12V
E30	GND
E31	12V
E32	12V
E33	GND
E34	12V
E35	12V
E36	GND
E37	12V
E38	12V
E39	GND
E40	12V

Column E

J1F

F1	GND
F2	GND
F3	GND
F4	GND
F5	GND
F6	GND
F7	PCIE_REFCLK_N
F8	GND
F9	GND
F10	GND
F11	GND
F12	GND
F13	GND
F14	GND
F15	GND
F16	GND
F17	GND
F18	PWR_BRAKE#
F19	GND
F20	JTAG_TDI
F21	GND
F22	12V
F23	12V
F24	GND
F25	12V
F26	12V
F27	GND
F28	12V
F29	12V
F30	GND
F31	12V
F32	12V
F33	GND
F34	12V
F35	12V
F36	GND
F37	12V
F38	12V
F39	GND
F40	12V

Column F

J1G

G1	PETp0
G2	PETn0
G3	GND
G4	PETp4
G5	PETn4
G6	GND
G7	GND
G8	GND
G9	PETp8
G10	PETn8
G11	GND
G12	PETp15
G13	PETn15
G14	GND
G15	GND
G16	UNRESOLVED/NC
G17	UNRESOLVED/NC
G18	GND
G19	GND
G20	JTAG_TDO
G21	GND
G22	12V
G23	12V
G24	GND
G25	12V
G26	12V
G27	GND
G28	12V
G29	12V
G30	GND
G31	12V
G32	12V
G33	GND
G34	12V
G35	12V
G36	GND
G37	12V
G38	12V
G39	GND
G40	12V

Column G

J1H

H1	GND
H2	PETp1
H3	PETn1
H4	GND
H5	PETp7
H6	PETn7
H7	GND
H8	PETp9
H9	PETn9
H10	GND
H11	PETp12
H12	PETn12
H13	GND
H14	GND
H15	UNRESOLVED/NC
H16	UNRESOLVED/NC
H17	GND
H18	I2C_ADDR1
H19	JTAG_SEL
H20	JTAG_TCLK
H21	GND
H22	12V
H23	12V
H24	GND
H25	12V
H26	12V
H27	GND
H28	12V
H29	12V
H30	GND
H31	12V
H32	12V
H33	GND
H34	12V
H35	12V
H36	GND
H37	12V
H38	12V
H39	GND
H40	12V

Column H

J1I

J1	PETp2
J2	PETn2
J3	GND
J4	PETp5
J5	PETn5
J6	GND
J7	GND
J8	GND
J9	PETp11
J10	PETn11
J11	GND
J12	PETp14
J13	PETn14
J14	GND
J15	GND
J16	UNRESOLVED/NC
J17	UNRESOLVED/NC
J18	GND
J19	GND
J20	JTAG_TMS
J21	GND
J22	12V
J23	12V
J24	GND
J25	12V
J26	12V
J27	GND
J28	12V
J29	12V
J30	GND
J31	12V
J32	12V
J33	GND
J34	12V
J35	12V
J36	GND
J37	12V
J38	12V
J39	GND
J40	12V

Column J

J1J

K1	GND
K2	PETp3
K3	PETn3
K4	GND
K5	PETp6
K6	PETn6
K7	GND
K8	PETp10
K9	PETn10
K10	GND
K11	PETp13
K12	PETn13
K13	GND
K14	GND
K15	UNRESOLVED/NC
K16	UNRESOLVED/NC
K17	GND
K18	PWR_EN
K19	THERM_OVERT#
K20	JTAG_TRST#
K21	GND
K22	12V
K23	12V
K24	GND
K25	12V
K26	12V
K27	GND
K28	12V
K29	12V
K30	GND
K31	12V
K32	12V
K33	GND
K34	12V
K35	12V
K36	GND
K37	12V
K38	12V
K39	GND
K40	12V

Column K

Canonical contact names only; footprint orientation is not assigned

Pigeon Labs

Sheet: /Connector A – 400 contacts/
File: Connector_A.kicad_sch

Title: V100 SXM2 – Connector A reference

Size: A3	Date: 2026-10-29	**Rev: Private review**
KiCad E.D.A. 10.0.7		Id: 2/3

All pins are exposed for reuse. Unconnected here does not mean an official NC assignment.

J2A

A1	NVHS3_TX_L5_N
A2	NVHS3_TX_L5_P
A3	GND
A4	NVHS3_TX_L2_N
A5	NVHS3_TX_L2_P
A6	GND
A7	NVHS2_TX_L5_N
A8	NVHS2_TX_L5_P
A9	GND
A10	NVHS2_TX_L2_N
A11	NVHS2_TX_L2_P
A12	GND
A13	GND
A14	GND
A15	NVHS5_TX_L4_N
A16	NVHS5_TX_L4_P
A17	GND
A18	NVHS5_TX_L1_N
A19	NVHS5_TX_L1_P
A20	GND
A21	NVHS4_TX_L5_N
A22	NVHS4_TX_L5_P
A23	GND
A24	NVHS4_TX_L2_N
A25	NVHS4_TX_L2_P
A26	GND
A27	GND
A28	GND
A29	NVHS1_TX_L4_N
A30	NVHS1_TX_L4_P
A31	GND
A32	NVHS1_TX_L1_N
A33	NVHS1_TX_L1_P
A34	GND
A35	NVHS0_TX_L5_N
A36	NVHS0_TX_L5_P
A37	GND
A38	NVHS0_TX_L2_N
A39	NVHS0_TX_L2_P
A40	GND

Column A

J2B

B1	GND
B2	NVHS3_TX_L4_N
B3	NVHS3_TX_L4_P
B4	GND
B5	NVHS3_TX_L1_N
B6	NVHS3_TX_L1_P
B7	GND
B8	NVHS2_TX_L4_N
B9	NVHS2_TX_L4_P
B10	GND
B11	NVHS2_TX_L1_N
B12	NVHS2_TX_L1_P
B13	GND
B14	NVHS5_TX_L5_N
B15	NVHS5_TX_L5_P
B16	GND
B17	NVHS5_TX_L2_N
B18	NVHS5_TX_L2_P
B19	GND
B20	GND
B21	GND
B22	NVHS4_TX_L4_N
B23	NVHS4_TX_L4_P
B24	GND
B25	NVHS4_TX_L1_N
B26	NVHS4_TX_L1_P
B27	GND
B28	NVHS1_TX_L5_N
B29	NVHS1_TX_L5_P
B30	GND
B31	NVHS1_TX_L2_N
B32	NVHS1_TX_L2_P
B33	GND
B34	GND
B35	GND
B36	NVHS0_TX_L4_N
B37	NVHS0_TX_L4_P
B38	GND
B39	NVHS0_TX_L1_N
B40	NVHS0_TX_L1_P

Column B

J2C

C1	NVHS3_TX_L6_N
C2	NVHS3_TX_L6_P
C3	GND
C4	NVHS3_TX_L3_N
C5	NVHS3_TX_L3_P
C6	GND
C7	NVHS2_TX_L6_N
C8	NVHS2_TX_L6_P
C9	GND
C10	NVHS2_TX_L3_N
C11	NVHS2_TX_L3_P
C12	GND
C13	GND
C14	GND
C15	NVHS5_TX_L6_N
C16	NVHS5_TX_L6_P
C17	GND
C18	NVHS5_TX_L0_N
C19	NVHS5_TX_L0_P
C20	GND
C21	NVHS4_TX_L6_N
C22	NVHS4_TX_L6_P
C23	GND
C24	NVHS4_TX_L3_N
C25	NVHS4_TX_L3_P
C26	GND
C27	GND
C28	GND
C29	NVHS1_TX_L6_N
C30	NVHS1_TX_L6_P
C31	GND
C32	NVHS1_TX_L0_N
C33	NVHS1_TX_L0_P
C34	GND
C35	NVHS0_TX_L6_N
C36	NVHS0_TX_L6_P
C37	GND
C38	NVHS0_TX_L3_N
C39	NVHS0_TX_L3_P
C40	GND

Column C

J2D

D1	GND
D2	NVHS3_TX_L7_N
D3	NVHS3_TX_L7_P
D4	GND
D5	NVHS3_TX_L0_N
D6	NVHS3_TX_L0_P
D7	GND
D8	NVHS2_TX_L7_N
D9	NVHS2_TX_L7_P
D10	GND
D11	NVHS2_TX_L0_N
D12	NVHS2_TX_L0_P
D13	GND
D14	NVHS5_TX_L7_N
D15	NVHS5_TX_L7_P
D16	GND
D17	NVHS5_TX_L3_N
D18	NVHS5_TX_L3_P
D19	GND
D20	GND
D21	GND
D22	NVHS4_TX_L7_N
D23	NVHS4_TX_L7_P
D24	GND
D25	NVHS4_TX_L0_N
D26	NVHS4_TX_L0_P
D27	GND
D28	NVHS1_TX_L7_N
D29	NVHS1_TX_L7_P
D30	GND
D31	NVHS1_TX_L3_N
D32	NVHS1_TX_L3_P
D33	GND
D34	GND
D35	GND
D36	NVHS0_TX_L7_N
D37	NVHS0_TX_L7_P
D38	GND
D39	NVHS0_TX_L0_N
D40	NVHS0_TX_L0_P

Column D

J2E

E1	GND
E2	GND
E3	GND
E4	GND
E5	GND
E6	GND
E7	GND
E8	GND
E9	GND
E10	GND
E11	GND
E12	GND
E13	GND
E14	GND
E15	GND
E16	GND
E17	GND
E18	GND
E19	GND
E20	NVHS_REFCLK_P
E21	GND
E22	GND
E23	GND
E24	GND
E25	GND
E26	GND
E27	GND
E28	GND
E29	GND
E30	GND
E31	GND
E32	GND
E33	GND
E34	SMB_CLK_(candidate)
E35	GND
E36	GND
E37	GND
E38	GND
E39	GND
E40	GND

Column E

J2F

F1	GND
F2	GND
F3	GND
F4	GND
F5	GND
F6	GND
F7	GND
F8	GND
F9	GND
F10	GND
F11	GND
F12	GND
F13	GND
F14	GND
F15	GND
F16	GND
F17	GND
F18	GND
F19	GND
F20	NVHS_REFCLK_N
F21	GND
F22	GND
F23	GND
F24	GND
F25	GND
F26	GND
F27	GND
F28	GND
F29	GND
F30	GND
F31	GND
F32	GND
F33	GND
F34	SMB_DAT_(candidate)
F35	GND
F36	GND
F37	GND
F38	GND
F39	GND
F40	GND

Column F

J2G

G1	GND
G2	NVHS3_RX_L4_N
G3	NVHS3_RX_L4_P
G4	GND
G5	NVHS3_RX_L1_N
G6	NVHS3_RX_L1_P
G7	GND
G8	NVHS2_RX_L4_N
G9	NVHS2_RX_L4_P
G10	GND
G11	NVHS2_RX_L1_N
G12	NVHS2_RX_L1_P
G13	GND
G14	NVHS5_RX_L5_N
G15	NVHS5_RX_L5_P
G16	GND
G17	NVHS5_RX_L2_N
G18	NVHS5_RX_L2_P
G19	GND
G20	GND
G21	GND
G22	NVHS4_RX_L4_N
G23	NVHS4_RX_L4_P
G24	GND
G25	NVHS4_RX_L1_N
G26	NVHS4_RX_L1_P
G27	GND
G28	NVHS1_RX_L5_N
G29	NVHS1_RX_L5_P
G30	GND
G31	NVHS1_RX_L2_N
G32	NVHS1_RX_L2_P
G33	GND
G34	GND
G35	GND
G36	NVHS0_RX_L4_N
G37	NVHS0_RX_L4_P
G38	GND
G39	NVHS0_RX_L1_N
G40	NVHS0_RX_L1_P

Column G

J2H

H1	NVHS3_RX_L5_N
H2	NVHS3_RX_L5_P
H3	GND
H4	NVHS3_RX_L2_N
H5	NVHS3_RX_L2_P
H6	GND
H7	NVHS2_RX_L5_N
H8	NVHS2_RX_L5_P
H9	GND
H10	NVHS2_RX_L2_N
H11	NVHS2_RX_L2_P
H12	GND
H13	PRSNT1_A_(candidate)
H14	GND
H15	NVHS5_RX_L4_N
H16	NVHS5_RX_L4_P
H17	GND
H18	NVHS5_RX_L1_N
H19	NVHS5_RX_L1_P
H20	GND
H21	NVHS4_RX_L5_N
H22	NVHS4_RX_L5_P
H23	GND
H24	NVHS4_RX_L2_N
H25	NVHS4_RX_L2_P
H26	GND
H27	GND
H28	GND
H29	NVHS1_RX_L4_N
H30	NVHS1_RX_L4_P
H31	GND
H32	NVHS1_RX_L1_N
H33	NVHS1_RX_L1_P
H34	GND
H35	NVHS0_RX_L5_N
H36	NVHS0_RX_L5_P
H37	GND
H38	NVHS0_RX_L2_N
H39	NVHS0_RX_L2_P
H40	GND

Column H

J2I

J1	GND
J2	NVHS3_RX_L6_N
J3	NVHS3_RX_L6_P
J4	GND
J5	NVHS3_RX_L0_N
J6	NVHS3_RX_L0_P
J7	GND
J8	NVHS2_RX_L6_N
J9	NVHS2_RX_L6_P
J10	GND
J11	NVHS2_RX_L0_N
J12	NVHS2_RX_L0_P
J13	GND
J14	NVHS5_RX_L6_N
J15	NVHS5_RX_L6_P
J16	GND
J17	NVHS5_RX_L3_N
J18	NVHS5_RX_L3_P
J19	GND
J20	GND
J21	GND
J22	NVHS4_RX_L6_N
J23	NVHS4_RX_L6_P
J24	GND
J25	NVHS4_RX_L0_N
J26	NVHS4_RX_L0_P
J27	GND
J28	NVHS1_RX_L6_N
J29	NVHS1_RX_L6_P
J30	GND
J31	NVHS1_RX_L3_N
J32	NVHS1_RX_L3_P
J33	GND
J34	PRSNT1_B_(candidate)
J35	GND
J36	NVHS0_RX_L6_N
J37	NVHS0_RX_L6_P
J38	GND
J39	NVHS0_RX_L0_N
J40	NVHS0_RX_L0_P

Column J

J2J

K1	NVHS3_RX_L7_N
K2	NVHS3_RX_L7_P
K3	GND
K4	NVHS3_RX_L3_N
K5	NVHS3_RX_L3_P
K6	GND
K7	NVHS2_RX_L7_N
K8	NVHS2_RX_L7_P
K9	GND
K10	NVHS2_RX_L3_N
K11	NVHS2_RX_L3_P
K12	GND
K13	GND
K14	GND
K15	NVHS5_RX_L7_N
K16	NVHS5_RX_L7_P
K17	GND
K18	NVHS5_RX_L0_N
K19	NVHS5_RX_L0_P
K20	GND
K21	NVHS4_RX_L7_N
K22	NVHS4_RX_L7_P
K23	GND
K24	NVHS4_RX_L3_N
K25	NVHS4_RX_L3_P
K26	GND
K27	GND
K28	GND
K29	NVHS1_RX_L7_N
K30	NVHS1_RX_L7_P
K31	GND
K32	NVHS1_RX_L0_N
K33	NVHS1_RX_L0_P
K34	GND
K35	NVHS0_RX_L7_N
K36	NVHS0_RX_L7_P
K37	GND
K38	NVHS0_RX_L3_N
K39	NVHS0_RX_L3_P
K40	GND

Column K

Canonical contact names only; footprint orientation is not assigned

Pigeon Labs

Sheet: /Connector B — 400 contacts/
File: Connector_B.kicad_sch

Title: V100 SXM2 — Connector B reference

Size: A3	Date: 2026-10-29	**Rev: Private review**
KiCad E.D.A. 10.0.7		Id: 3/3